Faculty of Computer Applications

INVERTIS UNIVERSITY BAREILLY

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Introduction To Data Analytics

Supply Chain Analytics



Master of Computer Applications

Submitted To:

Mr. Divyank Chauhan

Data Science Trainer

Submitted By:

Team Name: **02 MCA Sec. A**

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Total Member: 10

Invertis University, Bareilly [UP]

(2025-26)

DECLARATION

I, Dharmesh Kushwaha, hereby declare that the project entitled "Supply Chain Analytics Dashboard" submitted for the Master of Computer Applications (MCA) degree is my original work and has been carried out under the guidance of Mr. Divyank Chauhan, Assistant Professor, Department of Computer Applications.

The work embodied in this project has not been submitted elsewhere for the award of any other degree or diploma.

All sources of information have been duly acknowledged. The project work was completed as a collaborative effort with my team members, and each member contributed equally to its successful completion.

Team Members' Contribution:

S. No.	Student Name	Contribution
1	Dharmesh Kushwaha (Leader)	100%
2	Nitin Verma	100%
3	Kabir Moshin	100%
4	Ronit Maurya	100%
5	Mandeep Kumar	100%
6	Nishtha Agarwal	100%
7	Aditya Chauhan	100%
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10	Abhay Patel	40%

Place:

Date:

Signature of Team Leader MCA

Section A

.....

CERTIFICATE

This is to certify that the project entitled "Supply Chain Analytics Dashboard" is a bonafide work carried out by Dharmesh Kushwaha (Team Leader) along with the team members of MCA Section A, during the academic year 2026-2027, in partial fulfillment of the requirements for the award of Master of Computer Applications (MCA) degree.

The project work has been carried out under my guidance and supervision. The team demonstrated excellent teamwork, technical skills, and dedication throughout the project duration.

I am satisfied with the quality of work and recommend this project for evaluation.

_____

Mr. Divyank Chauhan

Data Science Trainer

Project Guide

External Examiner: _____

Date: _____

Signature: _____

ACKNOWLEDGMENT

We would like to express our sincere gratitude to all those who have contributed to the successful completion of this project on "Supply Chain Analytics Dashboard."

First and foremost, we extend our heartfelt thanks to our esteemed guide, Mr. Divyank Chauhan, Data Science Trainer, Department of Computer Applications, for his invaluable guidance, constant encouragement, and insightful suggestions throughout the project. His expertise and mentorship were instrumental in shaping this project.

We are deeply grateful to Dr. Akash Sanghi , Head of the Department of Computer Applications, for providing us with the necessary facilities and resources to undertake this project.

We would also like to thank our college management for creating an environment conducive to learning and research, which enabled us to explore innovative ideas and implement them effectively.

Our sincere appreciation goes to all faculty members of the Department of Computer Applications who directly or indirectly helped us during the course of this project with their valuable inputs and suggestions.

We are thankful to our family members and friends for their unwavering support and encouragement, which motivated us to put in our best efforts.

Last but not least, we acknowledge the collaborative spirit of our team members. Each team member contributed wholeheartedly, bringing diverse skills and perspectives that enriched the project. The synergy and coordination among the team made this project a rewarding learning experience.

We hope this project contributes to the field of supply chain data analytics and serves as a useful reference for future research.

Team Members:
Dharmesh Kushwaha, Nitin Verma, Kabir Moshin, Ronit Maurya, Mandeep Kumar, Nishtha Agarwal, Aditya Chauhan, Aasthik Mishra, Ayush Agarwal, Abhay Patel

MCA Section A

ABSTRACT

Project Title: Supply Chain Analytics Dashboard
Domain: Supply Chain Management, Data Analytics and Business Intelligence
Technology Stack: Microsoft Excel (structured tables, PivotTables, charts and formulas), Power BI (Power Query, DAX, interactive visuals), Python (pandas, matplotlib)

This project presents the development of a Supply Chain Analytics Dashboard using Microsoft Excel, Power BI and Python. The main objective of this project is to transform raw supply chain data into meaningful visual insights that support effective decision-making in procurement, inventory management, warehousing, transportation, and order delivery. In supply chain operations, data is often maintained across multiple records and tables, making it difficult to monitor inventory levels, evaluate supplier performance, track deliveries, analyze transportation costs, and assess overall operational efficiency.

To address these challenges, an interactive dashboard was first developed in Excel using Pivot Tables, Charts, Slicers, and KPI visualizations. The same data was then used to build an interactive Power BI dashboard with KPI cards, filters for warehouse, category and date, and visuals such as the monthly order trend, purchase value trend, inventory value, closing stock and delivery status. In addition, Python was used to analyze the data and create visualizations, including delivery status, customer type, sales by product, transportation cost by vehicle, and the relationship between distance and transportation cost. The project uses supply chain data related to suppliers, procurement, inventory, warehouses, sales orders, deliveries, and transportation costs.

The developed solution demonstrates how Microsoft Excel, Power BI and Python can be used together as effective data analytics and visualization tools for supply chain management. The project highlights how interactive dashboards can reduce manual reporting effort, improve data visibility, identify operational inefficiencies, and support data-driven decisions. Overall, the dashboards provide a consolidated view of important supply chain activities and help users understand performance through clear, organized, and interactive visualizations.

Keywords: Supply Chain Analytics, Excel Dashboard, Power BI, Python, Data Visualization, Pivot Tables, DAX, KPI, Inventory Management, Procurement, Transportation Cost, Delivery Performance.

TABLE OF CONTENT

CHAPTER 1: INTRODUCTION

1.1 Background .....
1.2 Motivation .....
1.3 Problem Statement .....
1.4 Objectives .....
1.5 Scope of the Project .....
1.6 Project Overview .....

CHAPTER 2: LITERATURE REVIEW

2.1 Existing System Analysis .....
2.2 Research Gap .....

CHAPTER 3: METHODOLOGY

3.1 Data Collection .....
3.2 Data Cleaning & Preprocessing .....
3.3 Data Analysis & Modeling …….....
3.4 Dashboard Designing Process .....

CHAPTER 4: SYSTEM DESIGN

4.1 Tools & Technologies Used .....
4.2 Data Flow Diagram .....
4.3 Architecture of Dashboard .....

CHAPTER 5: IMPLEMENTATION

5.1 KPI (Key Performance Indicator) Setup .....
5.2 Inventory Analysis .....
5.3 Procurement and Supplier Analysis .....
5.4 Sales and Order Analysis .....
5.5 Warehouse Analysis & Delivery Performance Analysis .....
5.6 Transportation Cost Analysis .....

CHAPTER 6: RESULTS & ANALYSIS

6.1 Dashboard Output Screenshots .....
6.2 Insights / Observations .....

CHAPTER 7: CONCLUSION & FUTURE SCOPE

7.1 Conclusion .....
7.2 Limitations .....
7.3 Future Scope .....

REFERENCES .....

ANNEXURES / APPENDIX (Raw Data & Code) .....

CHAPTER 1: INTRODUCTION

1.1 Background

In today's digital era, supply chain management generates a large amount of data related to suppliers, procurement, inventory, sales orders, warehouses, shipments, deliveries, and transportation costs. Organizations need to manage these activities efficiently to maintain product availability, control operational expenses, and ensure timely delivery of goods.

Data Analytics and Business Intelligence tools play an important role in transforming raw operational data into meaningful insights. A visual dashboard helps users monitor inventory levels, analyze procurement and sales activities, compare warehouse performance, and evaluate delivery outcomes.

This project focuses on developing a Supply Chain Analytics Dashboard using Microsoft Excel. The dashboard presents important supply chain information through charts, key performance indicators (KPIs), and analytical summaries, enabling users to understand operational performance and make informed decisions.

In addition to Excel, the project also uses Power BI to build an interactive dashboard and Python for further analysis and visualization of the same data.

1.2 Motivation

Supply chain data is often maintained across multiple worksheets and tables. Reviewing this information manually can be time-consuming and may make it difficult to identify important trends and operational issues.

This motivated the development of a dashboard that:

- Simplifies the interpretation of supply chain data.
- Reduces manual reporting effort.
- Provides meaningful visual insights into inventory, procurement, and sales.
- Helps analyze delivery performance and transportation costs.
- Extends the analysis through an interactive Power BI dashboard and Python-based visualizations.

Moreover, working on this project helped enhance practical skills in Microsoft Excel, data analysis, dashboard development, and data visualization, which are valuable in the data analytics industry. It also helped build practical skills in Power BI and Python.

1.3 Problem Statement

Supply chain operations generate information across multiple business activities, but analyzing these records separately makes it difficult to obtain a consolidated view of overall performance.

The major problems identified are:

- Data is distributed across multiple worksheets and tables.
- Inventory levels and stock values require systematic analysis.
- Procurement and sales activities are difficult to compare without summaries.
- Delivery performance and warehouse damage require additional analysis.
- Transportation expenses need to be summarized to understand logistics costs.
- Spreadsheet-based reports offer limited interactivity and limited support for deeper analysis.

Therefore, there is a need for an analytical dashboard that organizes supply chain data and presents important performance indicators in a clear and understandable format.

1.4 Objectives

The main objectives of this project are:

1. To organize supply chain data into structured tables for analysis.
2. To develop an analytical dashboard using Microsoft Excel.
3. To analyze inventory levels, stock value, and damaged quantities.
4. To examine procurement, sales orders, and supplier-related information.
5. To analyze delivery status, warehouse operations, and transportation costs.
6. To present important KPIs and visual summaries that support data-driven decision-making.
7. To develop an interactive dashboard using Power BI.
8. To use Python for further analysis and visualization of the supply chain data.

1.5 Scope of the Project

The scope of the project includes:

- Analyzing supplier, product, warehouse, and customer information.
- Examining procurement and sales order records.
- Analyzing inventory quantities, stock value, and damaged stock.
- Reviewing shipment, delivery, and transportation information.
- Creating dashboard visualizations and analytical summaries in Excel.
- Developing an interactive Power BI dashboard with filters for warehouse, category and date.
- Performing additional analysis and visualization using Python.

The dashboard can support students, data analysts, and supply chain management teams in understanding operational data and identifying areas that may require improvement.\

1.6 Project Overview

In this project, supply chain data stored in an Excel workbook was organized and analyzed using worksheets, supporting summaries, and dashboard visualizations. The workbook includes source data for suppliers, products, warehouses, customers, purchase orders, inventory, sales orders, shipments, deliveries, and transportation.

The analysis focuses on:

- Inventory quantity and stock value.
- Procurement and sales order analysis.
- Warehouse capacity and damaged stock.
- Delivery status and shipment performance.
- Transportation cost analysis.

The final dashboard consolidates important supply chain indicators and supporting visualizations into a single analytical view. It helps users review operational information more efficiently and understand patterns across different supply chain activities.

The same data was also used to create an interactive Power BI dashboard and a set of Python visualizations, such as delivery status, customer type, sales by product, transportation cost by vehicle, and distance versus transportation cost. These complement the Excel dashboard and allow the results to be compared across the three tools.

CHAPTER 2: LITERATURE REVIEW

2.1 Existing System Analysis

In many basic supply chain workflows, information about procurement, inventory, sales, warehouses, shipments, and deliveries is maintained in separate tables or spreadsheets. These records support routine operations, but analyzing them manually requires repeated filtering, sorting, and calculations.

When data is distributed across multiple worksheets, users may find it difficult to compare inventory values, monitor order statuses, evaluate warehouse damage, and understand transportation expenses. Preparing consolidated reports can also require additional time and effort.

Characteristics of the Existing System

Parameter	Existing System Description
Data Handling	Information is maintained in separate tables and worksheets.
Reporting	Reports may require manual filtering and calculations.
Data Analysis	Analysis depends on summaries and comparisons prepared by users.
Decision-Making	Important trends may be difficult to identify quickly.
Cost Monitoring	Transportation and inventory costs require consolidated analysis.

Problems Identified in the Existing System

1. Limited consolidated insights: Data from different supply chain activities must be reviewed separately.
2. Time-consuming reporting: Manual summaries require additional effort.
3. Inventory monitoring challenges: Stock quantities, stock values, and damaged quantities need systematic analysis.
4. Difficult performance comparison: Comparing procurement, sales, delivery, and warehouse information can be complicated.
5. Limited cost visibility: Transportation expenses may be distributed across different records and components.

These challenges demonstrate the importance of a structured analytical approach that brings key supply chain information together for easier interpretation.

2.2 Research Gap

Existing spreadsheet-based systems are useful for recording and organizing operational information. However, storing data alone does not automatically provide a clear overview of

business performance. Users need summarized indicators and visual representations to interpret large amounts of information effectively.

Existing Systems and Identified Gaps

Existing System / Tool	Identified Gap
Basic Excel worksheets	Information requires manual review and comparison.
Separate operational tables	Related data is not always visible in one consolidated view.
Manual reporting	Repeated calculations increase reporting effort.
Inventory records	Stock value and damaged quantities require analytical summaries.
Separate logistics records	Delivery status and transportation costs require combined analysis.
Excel dashboards	Limited interactivity and limited support for deeper analysis; addressed in this project using Power BI and Python.

The key need identified is a dashboard that converts operational records into useful summaries and visual insights.

The proposed dashboard addresses this need by presenting inventory, procurement, sales, warehouse, delivery, and transportation information in an organized format. It provides a foundation for understanding supply chain performance and identifying areas for further investigation.

The project also extends the Excel dashboard by using Power BI, which provides interactive filters and an interactive dashboard layout, and Python, which supports further analysis and visualization of relationships in the data, such as distance versus transportation cost.

CHAPTER 3: METHODOLOGY

3.1 Data Collection

The primary data source for this project was the supplied Excel workbook, which contains information related to different supply chain activities. The workbook includes master data, transaction records, and supporting analytical worksheets.

The dataset contains the following major categories:

- Supplier information
- Product information
- Warehouse information
- Customer information
- Purchase orders and purchase order items
- Inventory records
- Sales orders and sales order items
- Shipment and delivery records
- Transportation information

These datasets provide the foundation for analyzing inventory, procurement, sales, warehouse operations, delivery outcomes, and transportation expenses.

The same workbook was also used as the data source for the Power BI dashboard and for the Python analysis.

3.2 Data Cleaning and Preprocessing

Data preparation is an important stage of analytics because accurate and consistent information is necessary for reliable analysis. The workbook was reviewed to understand its structure, identify relevant fields, and organize the data for analysis.

The main data preparation activities include:

Step	Purpose
Data Inspection	Review worksheet structures and available columns.
Data Formatting	Ensure that numerical values, dates, and text fields are appropriately represented.
Data Consistency	Review category names, status values, and field formats.
Data Organization	Identify relevant source tables for each analytical requirement.
Data Validation	Check KPI definitions and calculations before interpreting results.
Data Loading (Power BI)	Load the Excel tables into Power BI and set the data types and relationships.

Step	Purpose
Data Checking (Python)	Read the workbook in Python and check totals and category counts against the Excel and Power BI values.

These activities help improve the clarity of the dataset and support the creation of meaningful analytical summaries.

3.3 Data Modeling(Pivot Tables / Excel Analysis)

Data modeling and summarization organize supply chain information into a form suitable for analysis. In this project, source tables and supporting pivot-style summaries were used to examine different operational measures.

The analytical activities include:

1. Summarizing inventory quantities, stock values, and damaged quantities.
2. Analyzing procurement totals and sales order information.
3. Comparing warehouse-related measures.
4. Reviewing delivery statuses and transportation expenses.
5. Organizing analytical results for presentation in the dashboard.
6. Creating DAX measures in Power BI, such as Total Purchase Value, Total Sales Value, Inventory Value and Transport Cost, for the dashboard KPIs.
7. Using Python to summarize the data by category, warehouse, vehicle and month, and to study relationships such as distance versus transportation cost and closing stock versus damaged quantity.

The supporting summaries act as the analytical foundation for the dashboard and help users compare different categories and operational measures.

3.4 Dashboard Designing Process

After identifying the relevant data and analytical measures, the dashboard was organized to present important indicators and visual summaries in a clear layout.

The dashboard design process includes:

Step	Description
KPI Selection	Identify important measures related to supply chain performance.
Chart Selection	Choose suitable visualizations for inventory, orders, deliveries, and costs.
Data Summarization	Use supporting analytical worksheets to prepare chart data.

Step	Description
Layout Design	Arrange indicators and charts in a structured format.
Visual Presentation	Use appropriate labels, formatting, and chart titles to improve readability.
Interactive Filters (Power BI)	Add slicers for warehouse, category and date so that users can filter all visuals.
Theme and Background (Power BI)	Apply a supply chain themed background, consistent colours and a centred title.
Python Visualization	Create additional charts in Python to show relationships not covered in the dashboards.

The final dashboard provides a consolidated view of important supply chain measures and helps users interpret the information more efficiently.

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CHAPTER 4: SYSTEM DESIGN

4.1 Tools and Technologies Used

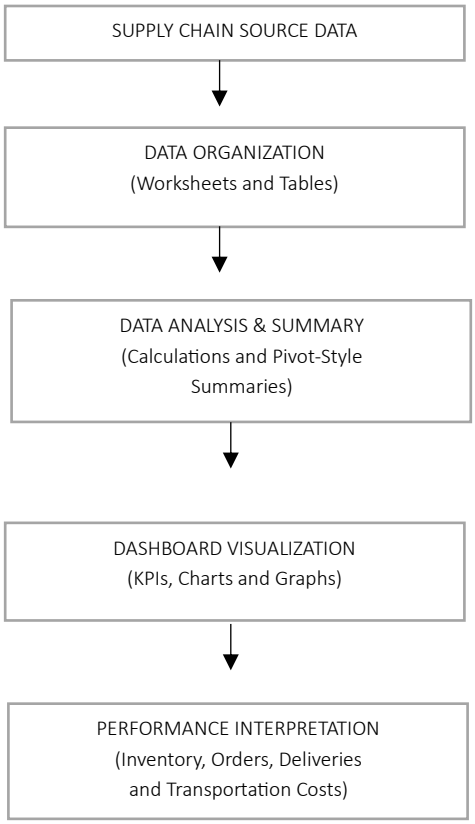
The project was developed using Microsoft Excel and its spreadsheet-based analysis and visualization capabilities. Power BI and Python were also used for the interactive dashboard and for further analysis.

Tool / Technology	Purpose / Function
Microsoft Excel	Main platform for data organization, analysis, and dashboard development.
Excel Worksheets	Store and organize source data and analytical summaries.
Pivot Tables / Summary Sheets	Summarize and compare supply chain information.
Charts and Graphs	Present inventory, order, delivery, and cost information visually.
Excel Formulas	Support calculations and key performance indicators.
Cell Formatting	Improve the readability and presentation of the dashboard.

Microsoft Excel was selected because it provides accessible tools for organizing data, performing calculations, creating charts, and presenting analytical results without requiring a dedicated business intelligence platform.

4.2 Data Flow Diagram (DFD)

The project follows a workflow that transforms source data into an organized dashboard.



Process Explanation

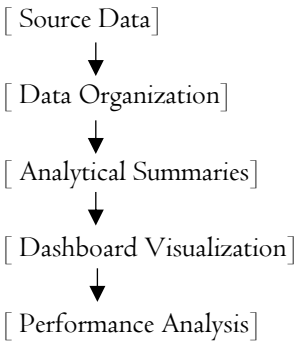
1. Supply chain information is obtained from the source worksheets.
2. Relevant tables and fields are identified and organized.
3. Data is summarized to calculate or compare important measures.
4. Charts and KPIs present the analytical results.
5. The results are interpreted to understand supply chain performance.

4.3 Architecture of Dashboard

The dashboard architecture consists of three main layers.

Layer	Component	Function
Data Layer	Excel source worksheets	Store supplier, product, inventory, order, delivery, and transportation records.
Processing Layer	Calculations and analytical summaries	Organize data and prepare the measures needed for analysis.
Visualization Layer	Dashboard charts and KPIs	Present summarized information in an understandable format.

Dashboard Architecture Flow



This structure separates source records from analytical results and makes it easier to understand how the dashboard converts data into useful information.

CHAPTER 5: IMPLEMENTATION

5.1 KPI (Key Performance Indicator) Setup

Key Performance Indicators help users understand important supply chain measures without reviewing every source worksheet individually.
The project focuses on the following measures:

KPI Name	Purpose
Total Purchase Value	Summarizes the value recorded in purchase orders.
Total Sales Value	Summarizes the value recorded in sales order items.
Total Inventory Value	Represents the recorded value of inventory stock.
Total Closing Stock	Shows the total recorded closing stock quantity.
Damaged Quantity	Indicates the total quantity of damaged inventory.
Transportation Cost	Summarizes recorded transportation expenses.
Delivery Status	Helps review the distribution of delivery outcomes.

These measures provide an overview of procurement, sales, inventory, logistics, and delivery activities. KPI calculations should be checked against their corresponding source columns before the results are used for decision-making.

5.2 Inventory Analysis Chart

Inventory analysis helps users understand the quantity and value of products maintained across the supply chain.
The analysis focuses on:

- Closing stock quantity.
- Inventory stock value.
- Damaged inventory quantity.
- Product or category-level stock information.

Charts and summaries can help identify categories with higher stock values and highlight damaged inventory that may require further investigation.
This analysis supports inventory monitoring and helps users understand how stock is distributed across the available records.

5.3 Procurement and Supplier Analysis

Procurement analysis examines purchase order information and supplier-related records. It helps users understand purchasing activity and review the recorded value of procurement transactions. The analysis includes:

- Total purchase order value.
- Purchase order status.
- Supplier information and recorded ratings.
- Product-level purchase details.

The resulting summaries help organize procurement information and provide a basis for comparing purchasing activity across suppliers, products, and order categories.

5.4 Sales and Order Analysis

Sales and order analysis focuses on customer orders and their associated line-item information. The main areas of analysis include:

- Sales order records.
- Sales line-item values.
- Order status distribution.
- Customer and product-related information.

These summaries help users review sales activity and understand the distribution of order records. Purchase order totals and sales line-item totals should be treated as separate measures because they come from different tables and do not directly establish profit.

5.5 Warehouse and Delivery Analysis

Warehouse and delivery analysis examines storage-related information and delivery outcomes. Warehouse analysis includes capacity, warehouse type, status, and damaged stock where those fields are available. Delivery analysis focuses on delivery status, delays, and related shipment information.

These visual summaries help users compare warehouse records and understand delivery outcomes. They can also highlight areas that may require further investigation, such as damaged inventory or delayed deliveries.

5.6 Transportation Cost Analysis

Transportation cost analysis examines recorded logistics expenses associated with moving goods through the supply chain.

The analysis focuses on:

- Total transportation cost.
- Transportation cost components, where available.
- Shipment-related information.
- Comparisons across recorded transportation categories.

The purpose of this analysis is to make transportation expenses easier to review and compare. Further analysis could examine cost per shipment or cost per unit of distance, provided the necessary data is available and the calculation is defined correctly.

CHAPTER 6: RESULTS AND ANALYSIS

6.1 Dashboard Output Screenshots

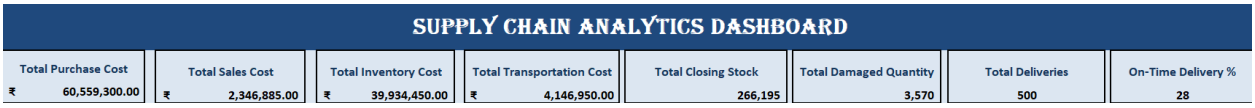
The completed Excel workbook contains a Final Dashboard sheet and supporting analysis worksheets. The dashboard presents important supply chain performance indicators through KPI cards, charts, and visual summaries. It helps users understand inventory levels, procurement and sales costs, warehouse performance, delivery status, transportation expenses, and customer distribution.

The dashboard combines different aspects of supply chain operations into a single visual interface. This makes it easier to compare operational measures and identify patterns in the available data. The supporting worksheets provide the data and analysis used to prepare the dashboard visualizations.

The following sections explain selected charts from the dashboard.

(A) Key Performance Indicators (KPIs)

Figure 6.1: Key Performance Indicators of the Supply Chain Analytics Dashboard



The dashboard displays eight key performance indicators (KPIs) that provide a summary of important supply chain activities. These indicators cover purchasing, sales, inventory, transportation, stock, damaged goods, deliveries, and on-time delivery performance.

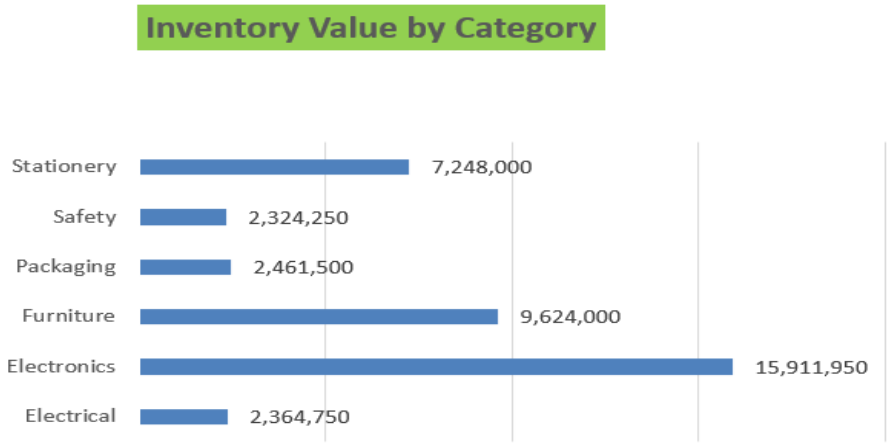
The KPI values displayed in the dashboard are summarized below:

KPI	Displayed Value
Total Purchase Cost	₹60,559,300.00
Total Sales Cost	₹2,346,885.00
Total Inventory Cost	₹39,934,450.00
Total Transportation Cost	₹4,146,950.00
Total Closing Stock	266,195
Total Damaged Quantity	3,570
Total Deliveries	500
On-Time Delivery %	28

These indicators help users quickly review the overall supply chain data without examining each record individually. Purchase, sales, inventory, and transportation costs provide financial information, while closing stock and damaged quantity summarize inventory-related measures. Total deliveries and the on-time delivery measure provide an overview of delivery operations.

(B) Inventory Analysis

Figure 6.2: Inventory Value by Category



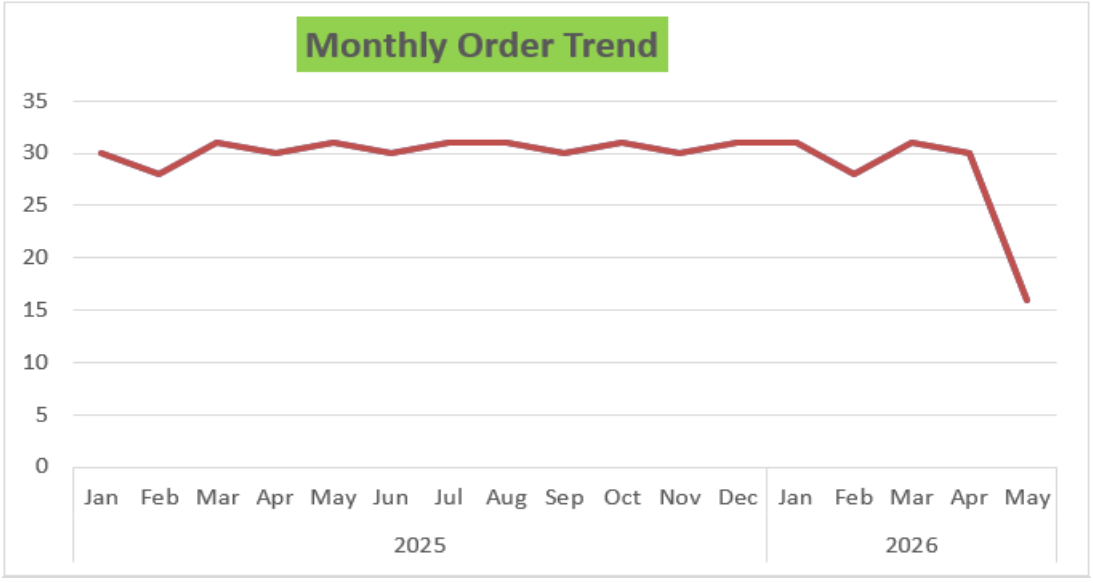
The Inventory Value by Category chart presents the distribution of inventory value across different product categories. It helps users compare the value of inventory held in each category and understand how inventory value is distributed throughout the supply chain.

This analysis can help identify categories that account for a relatively large or small share of total inventory value. Such information may support inventory planning, stock monitoring, and decisions regarding the allocation of resources.

The chart also provides a visual basis for comparing product categories without examining individual records separately. Actual conclusions about the highest-value category should be drawn from the values displayed in the chart.

(C) Monthly Order Trend Analysis

Figure 6.3: Monthly Order Trend



The Monthly Order Trend chart displays changes in order activity over the months represented in the dataset, including the available records for 2025 and 2026. It helps users examine variations in order volume over time.

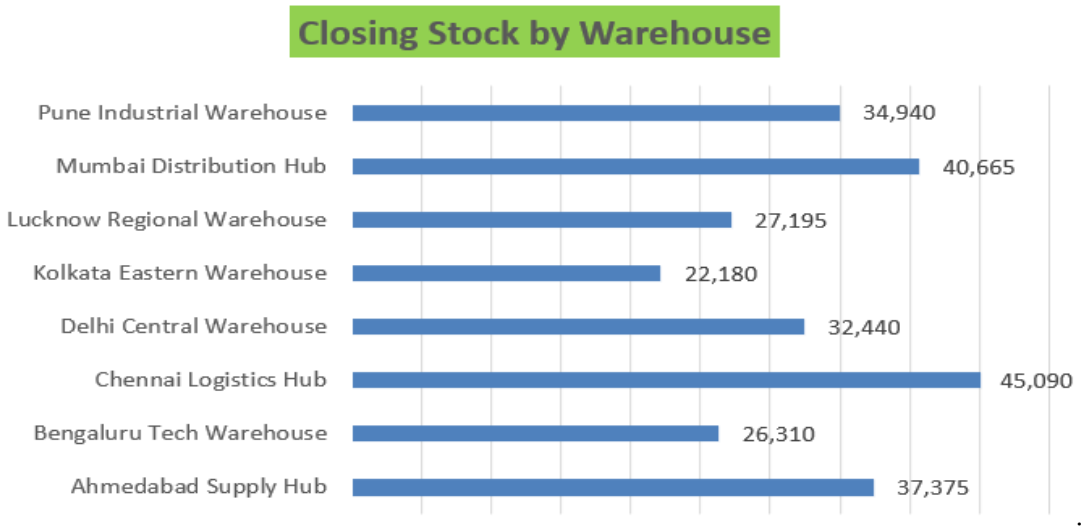
By comparing monthly values, users can identify periods with relatively high or low order activity. This information may be useful for planning inventory requirements, estimating procurement needs, and coordinating warehouse and transportation activities.

The chart also provides a basis for comparing order patterns across the two years. However, any conclusion about growth or decline should be based on the actual values shown in the visualization and the completeness of the available data.

(D) Warehouse Analysis

Closing Stock by Warehouse

Figure 6.4: Closing Stock by Warehouse



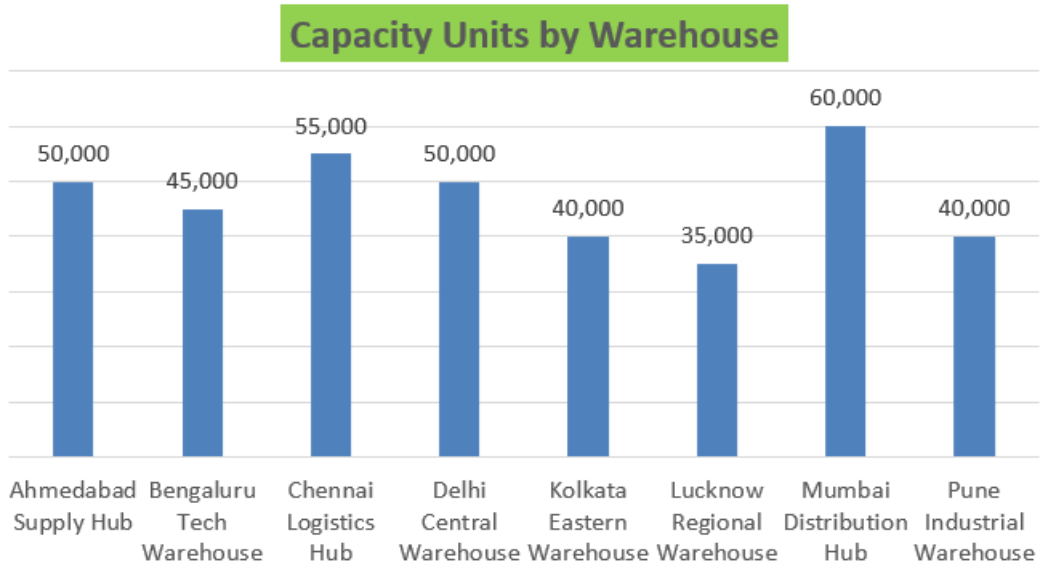
The Closing Stock by Warehouse chart compares the remaining inventory recorded at different warehouses. It helps users understand how stock is distributed among storage locations.

Comparing closing stock across warehouses can support inventory monitoring and stock allocation decisions. It may also help identify warehouses holding relatively high or low quantities of stock, which can be investigated further when planning transfers or replenishment.

The chart provides a summary of warehouse-level stock distribution and supports better visibility into inventory management.

Capacity Units by Warehouse

Figure 6.5: Capacity Units by Warehouse



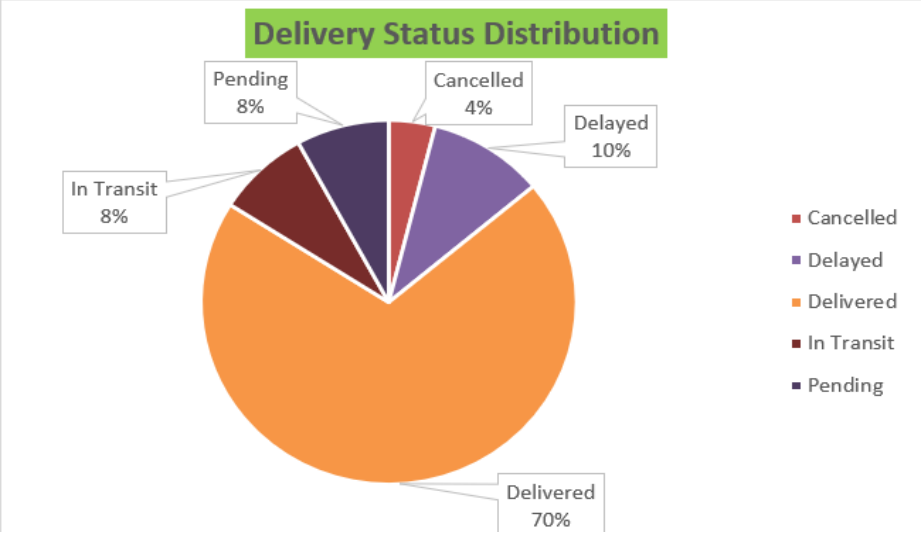
The Capacity Units by Warehouse chart presents the capacity information available for different warehouses. It allows users to compare the capacity recorded for each storage location.

When considered alongside the Closing Stock by Warehouse chart, this visualization helps users assess warehouse utilization more effectively. Warehouses with high closing stock relative to their available capacity may require closer monitoring, while locations with comparatively lower stock may have more room for additional inventory.

A detailed utilization assessment would require comparing stock quantity and usable warehouse capacity on a consistent basis. Therefore, the chart should be interpreted according to the units and measures defined in the workbook.

(E) Delivery Status Analysis

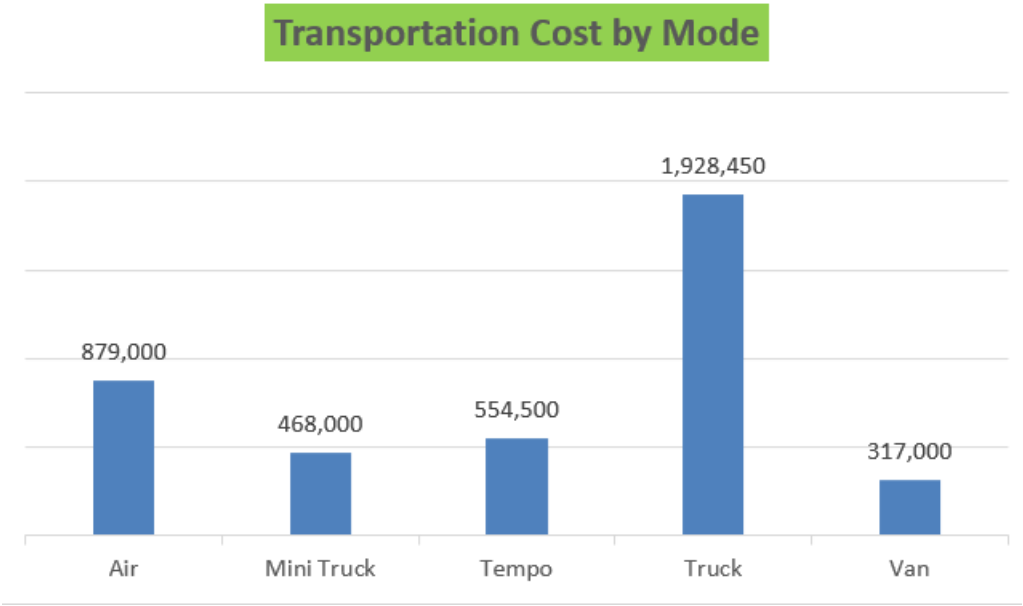
Figure 6.6: Delivery Status Distribution



The Delivery Status Distribution chart shows how deliveries are divided among the statuses recorded in the dataset, such as Delivered, Delayed, In Transit, Pending, and Cancelled.
This visualization provides an overview of delivery progress and helps users understand the distribution of completed and outstanding deliveries. It can also highlight the presence of delayed or cancelled deliveries that may require further investigation.
Monitoring delivery status is important for evaluating logistics activities and identifying areas where delivery coordination may need improvement. The chart can support operational reviews by making the distribution of delivery outcomes easier to understand.
The displayed percentages or quantities should be used when discussing the proportion of each delivery status.

(F) Transportation Cost Analysis

Figure 6.7: Transportation Cost by Mode

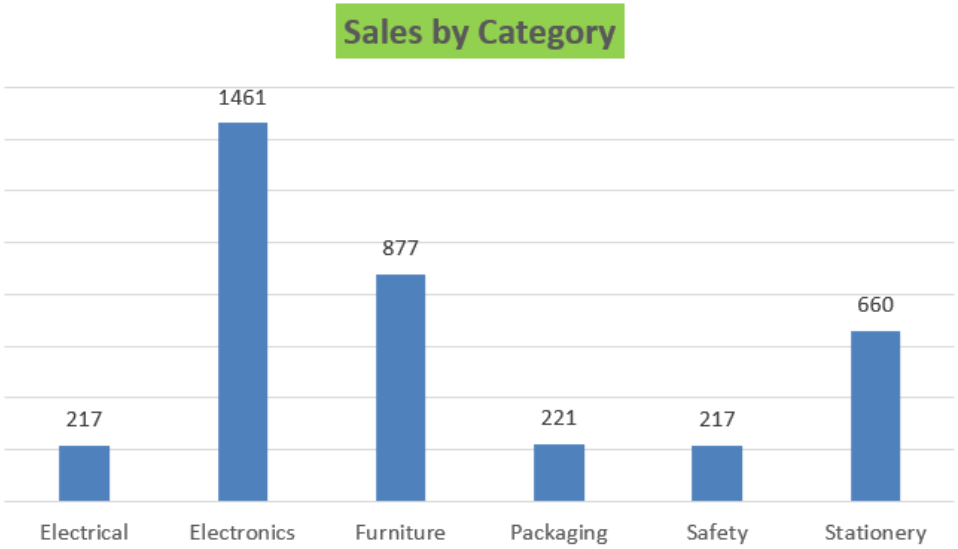


The Transportation Cost by Mode chart compares transportation expenses across the modes included in the dataset, such as Air, Mini Truck, Tempo, Truck, and Van.
This chart helps users examine how transportation costs vary across different modes. The comparison can support discussions about logistics expenses and the selection of suitable transportation options.
Transportation costs may vary because of factors such as distance, shipment size, vehicle type, and delivery requirements. However, the chart alone does not establish the reason for a cost difference. Additional information would be needed to determine whether a transportation mode is more cost-effective for a particular shipment.
Overall, the visualization improves the visibility of transportation expenditure and supports further cost analysis.

(G) Sales and Customer Analysis

Sales by Category

Figure 6.8: Sales by Category



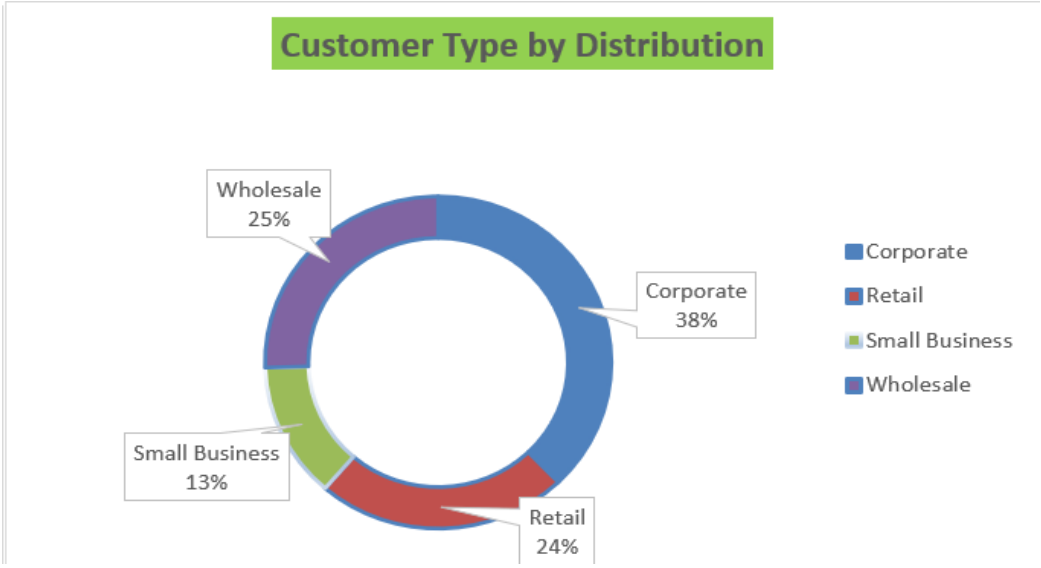
The Sales by Category chart presents sales information across the product categories represented in the dataset. It helps users compare the contribution of different categories to overall sales.

The visualization can help identify categories with relatively high or low recorded sales. This information may support sales monitoring, demand assessment, and coordination between sales and inventory planning.

Comparing sales by category with inventory value may also provide useful context for reviewing stock allocation. However, sales value and inventory value represent different measures and should not be treated as directly equivalent.

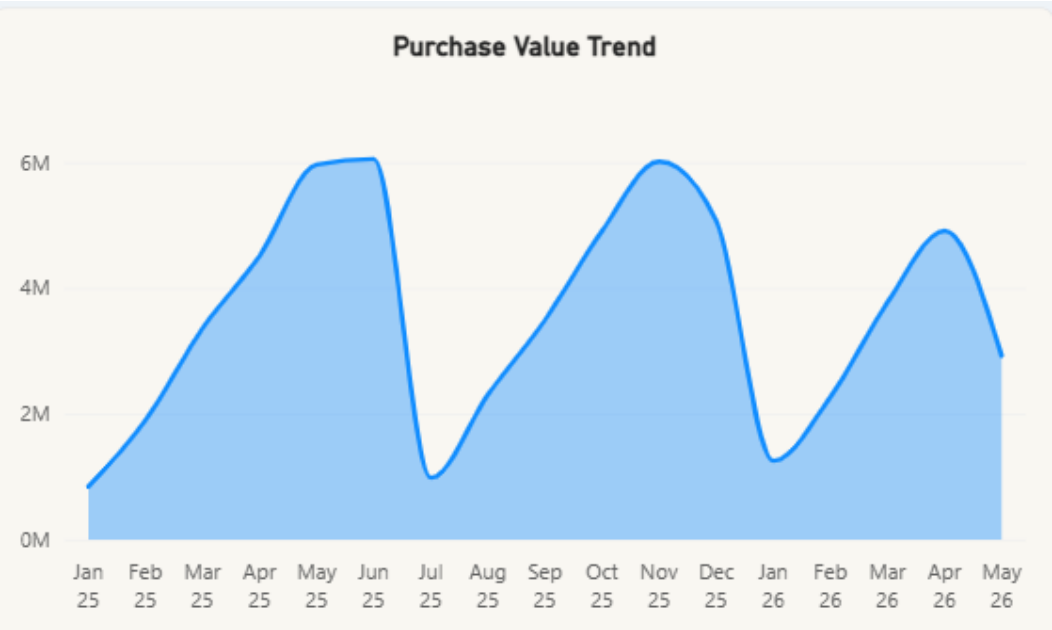
Customer Type Distribution

Figure 6.9: Customer Type Distribution



The Customer Type Distribution chart shows how customers are distributed across the customer types included in the dataset, such as Corporate, Retail, Wholesale, and Small Business.
This visualization helps users understand the composition of the customer base and compare the relative contribution of each customer type to the recorded distribution.
Customer classification can support supply chain planning by helping organizations understand the types of customers served. When combined with sales information, it may also help users investigate differences in purchasing patterns among customer groups.
The chart should be interpreted according to the measure used in the workbook, such as customer count, order count, or percentage distribution.

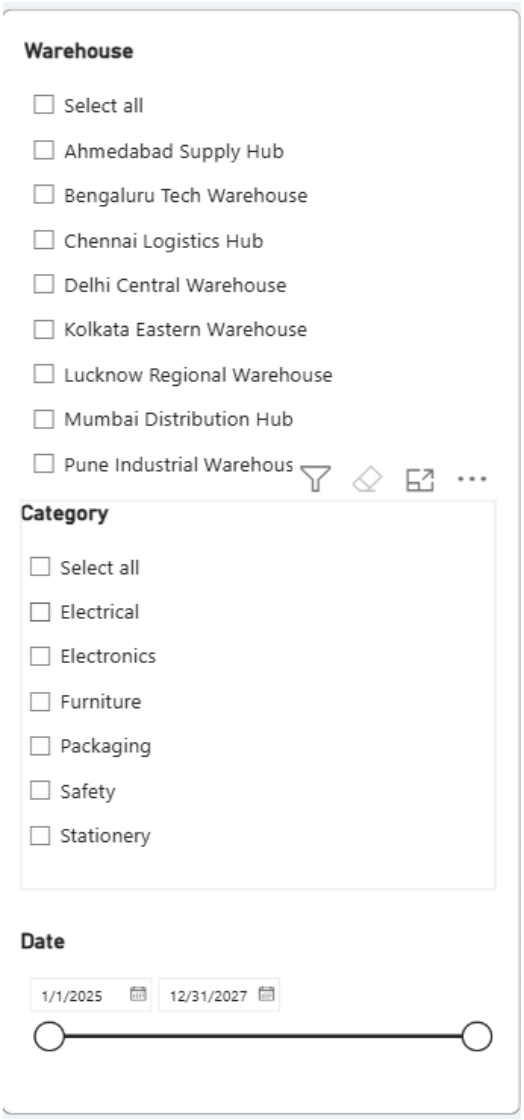
(H) Power BI Analysis
Purchase Value Trend
Figure 6.10: Purchase Value Trend



The Purchase Value Trend area chart shows the monthly purchase value from January 2025 to May 2026. The values rise and fall in repeating cycles, with peaks around May–June 2025, November 2025 and April 2026, and lower values around July 2025 and January 2026. This helps users identify periods of high and low purchasing activity and plan procurement and inventory accordingly. The most recent month should be interpreted according to the completeness of the available data.

Filters

Figure 6.11: Power BI Filters (Slicers)

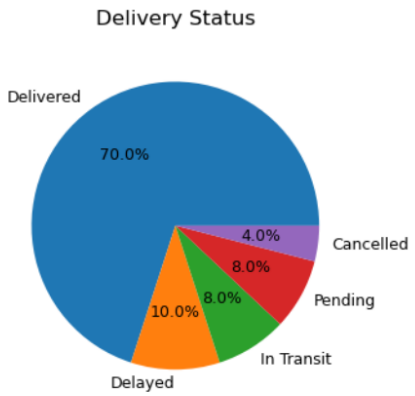


The Power BI dashboard includes filters for Warehouse, Category and Date. The Warehouse filter lists all eight warehouses, the Category filter lists Electrical, Electronics, Furniture, Packaging, Safety and Stationery, and the Date filter selects a date range. When a selection is made, all KPI cards and charts update at once, so users can review the data for a specific warehouse, category or period without changing the workbook.

(I) Python Analysis

Delivery Status

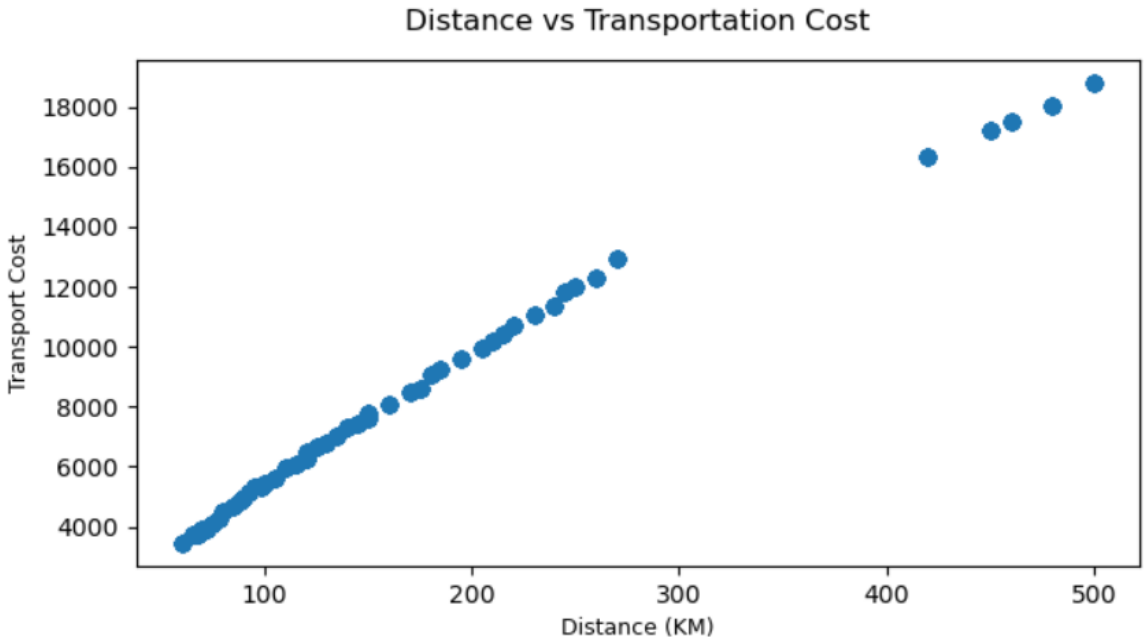
Figure 6.12: Delivery Status



The pie chart shows that 70% of deliveries were delivered, 10% delayed, 8% in transit, 8% pending and 4% cancelled. This gives a quick view of delivery progress and highlights the share of delayed and cancelled deliveries.

Distance vs Transportation Cost

Figure 6.13: Distance vs Transportation Cost (Python Scatter Plot)



The scatter plot shows a strong positive, almost linear relationship between distance and transportation cost: as distance increases, transportation cost increases. Distances range from about 60 km to 500 km, and there are no records between about 270 km and 420 km. This also supports the cost-per-distance analysis mentioned in Section 5.6.

6.2 Insights / Observations

The dashboards provide a consolidated view of several important supply chain activities. The following observations can be made from the analysis:

1. **Inventory visibility:** The Inventory Value by Category and Closing Stock by Warehouse charts help users examine inventory distribution across product categories and warehouse locations.
2. **Order monitoring:** The Monthly Order Trend chart makes it easier to review changes in monthly order activity and compare the periods represented in the dataset.
3. **Warehouse management:** The Closing Stock and Capacity Units charts provide complementary information for reviewing warehouse stock distribution and capacity.
4. **Delivery monitoring:** The Delivery Status Distribution chart summarizes delivery outcomes and helps identify statuses that may require additional attention.
5. **Transportation expenditure:** The Transportation Cost by Mode chart enables comparisons of recorded transportation expenses across the available modes.
6. **Sales and customer insights:** The Sales by Category and Customer Type Distribution charts help users examine sales distribution and understand the composition of the customer base.
7. **Performance overview:** The KPI cards summarize key measures, including purchase cost, sales cost, inventory cost, transportation cost, closing stock, damaged quantity, total deliveries, and the displayed on-time delivery measure.
8. **Purchase value trends (Power BI):** The Power BI Area Chart presents monthly purchase value trends, making it easier to identify periods of higher and lower purchasing activity. Its slicers also allow users to examine the dashboard according to warehouse, category, and date selections.
9. **Additional analysis using Python:** Python visualizations provide further insights into delivery status and transportation activities. The delivery status pie chart shows the distribution of delivery outcomes, while the scatter plot helps examine the relationship between transportation distance and transportation cost. The closing stock versus damaged quantity plot provides an additional view of the relationship between these two variables.

These observations describe the analytical capabilities of the dashboards. Specific claims about the best-performing warehouse, highest-value category, largest transportation expense, or overall delivery efficiency should be made only after checking the corresponding values in the workbook.

Key Takeaways

- The project combines Excel, Power BI, and Python to analyze different aspects of supply chain operations.
- The dashboards consolidate important supply chain information into analytical views.
- Supporting summaries and visualizations reduce the need to review every source table individually.
- Inventory, order, delivery, and transportation analysis can help identify areas requiring further investigation.
- Python and Power BI provide additional analytical and visualization capabilities alongside the Excel dashboard.

KPI definitions and calculations should be validated before drawing operational conclusions.

CHAPTER 7: CONCLUSION AND FUTURE SCOPE

7.1 Conclusion

The Supply Chain Analytics project demonstrates how operational data can be organized, analyzed, and presented through different data analytics and visualization tools, including Microsoft Excel, Python, and Power BI.

The project covers multiple supply chain activities, including supplier management, procurement, inventory, sales orders, warehouses, shipments, deliveries, and transportation costs. The data was organized and analyzed to understand different aspects of supply chain operations and present meaningful information through visualizations and analytical summaries.

Microsoft Excel was used for data organization, calculations, chart development, and dashboard creation. Python was used for additional data analysis and visualization, including delivery status distribution and the relationship between transportation distance and transportation cost. Power BI was used to develop an interactive dashboard with KPI cards, charts, and slicers for reviewing purchase value trends and other supply chain information.

Working on this project provided practical experience in data cleaning, exploratory data analysis, data visualization, dashboard development, and analytical reporting. It also highlighted the importance of data consistency and validating KPI definitions to ensure that the results are interpreted correctly.

Overall, the project demonstrates the practical application of Excel, Python, and Power BI in supply chain data analysis. It provides a foundation for understanding operational performance and can be extended through automation, database integration, and advanced analytical techniques.

7.2 Limitations

Although the project provides useful analytical insights, some limitations should be considered.

Limitation Area	Description
Data Validation	The accuracy of the analysis depends on the quality and consistency of the source records.
KPI Definitions	Some measures require clearly documented formulas and validation to ensure accurate interpretation.
Automation	Data refreshing and updating procedures depend on how the Excel workbook and Power BI reports have been configured.
Scalability	Excel-based analysis may become less convenient as the volume and complexity of data increase.

Advanced Analytics	The current project mainly focuses on descriptive analysis and visualization rather than advanced predictive modeling.
Data Availability	The analysis is limited to the information available in the supplied dataset.

The project should be reviewed and validated before its results are used for actual business operations.

7.3 Future Scope

The project can be further improved by introducing additional features and analytical techniques.

Enhancement	Description
Automated Data Refresh	Introduce repeatable data-refresh procedures to reduce manual updating efforts.
Advanced Power BI Reporting	Extend the existing dashboard with additional KPIs, analytical measures, and reporting capabilities.
Predictive Analytics	Explore demand forecasting, stock prediction, and transportation cost forecasting using machine learning techniques.
Inventory Alerts	Develop a system to identify products approaching their reorder levels or minimum stock limits.
Advanced Delivery Analysis	Calculate delivery performance using validated expected and actual delivery dates.
Database Integration	Connect the analytical system to a structured database for managing and analyzing larger datasets.
Advanced Python Analysis	Apply statistical techniques and machine learning models to identify patterns and improve forecasting.

These enhancements could improve the project's scalability and analytical capabilities while supporting more detailed supply chain decision-making.

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Note: Modify or add more references based on actual sources you used.

ANNEXURES / APPENDIX

Appendix A – Workbook Tables

The Excel workbook contains source tables and supporting worksheets related to the following supply chain activities:

- Suppliers
- Products
- Warehouses
- Customers
- Purchase Orders and Purchase Order Items
- Inventory
- Sales Orders and Sales Order Items
- Shipments and Deliveries
- Transportation

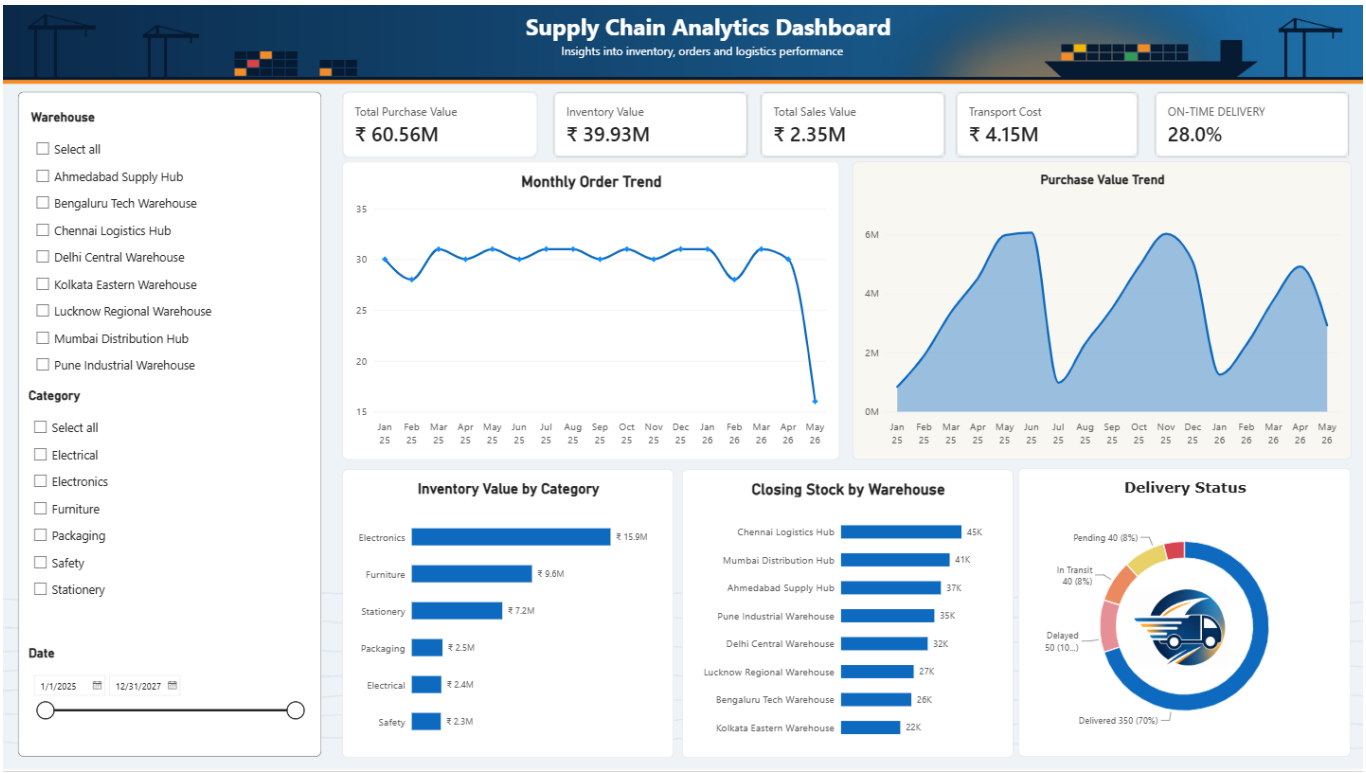
Appendix B – KPI Definitions

KPI	Definition
Total Purchase Value	Sum of the purchase order total_amount field.
Total Sales Value	Sum of the sales order item total_amount field.
Total Inventory Value	Sum of the inventory stock_value field.
Total Closing Stock	Sum of the inventory closing_stock field.
Total Damaged Quantity	Sum of the inventory damaged_qty field.
Total Transportation Cost	Sum of the transportation total_transport_cost field.
Delivered Share	Number of delivery records marked Delivered divided by the total number of delivery records, multiplied by 100.

Appendix C – Power BI Dashboard

This appendix presents the interactive Power BI dashboard developed using the supply chain dataset. It includes KPI cards, charts, an area chart for purchase value trends, and slicers for warehouse, category, and date.

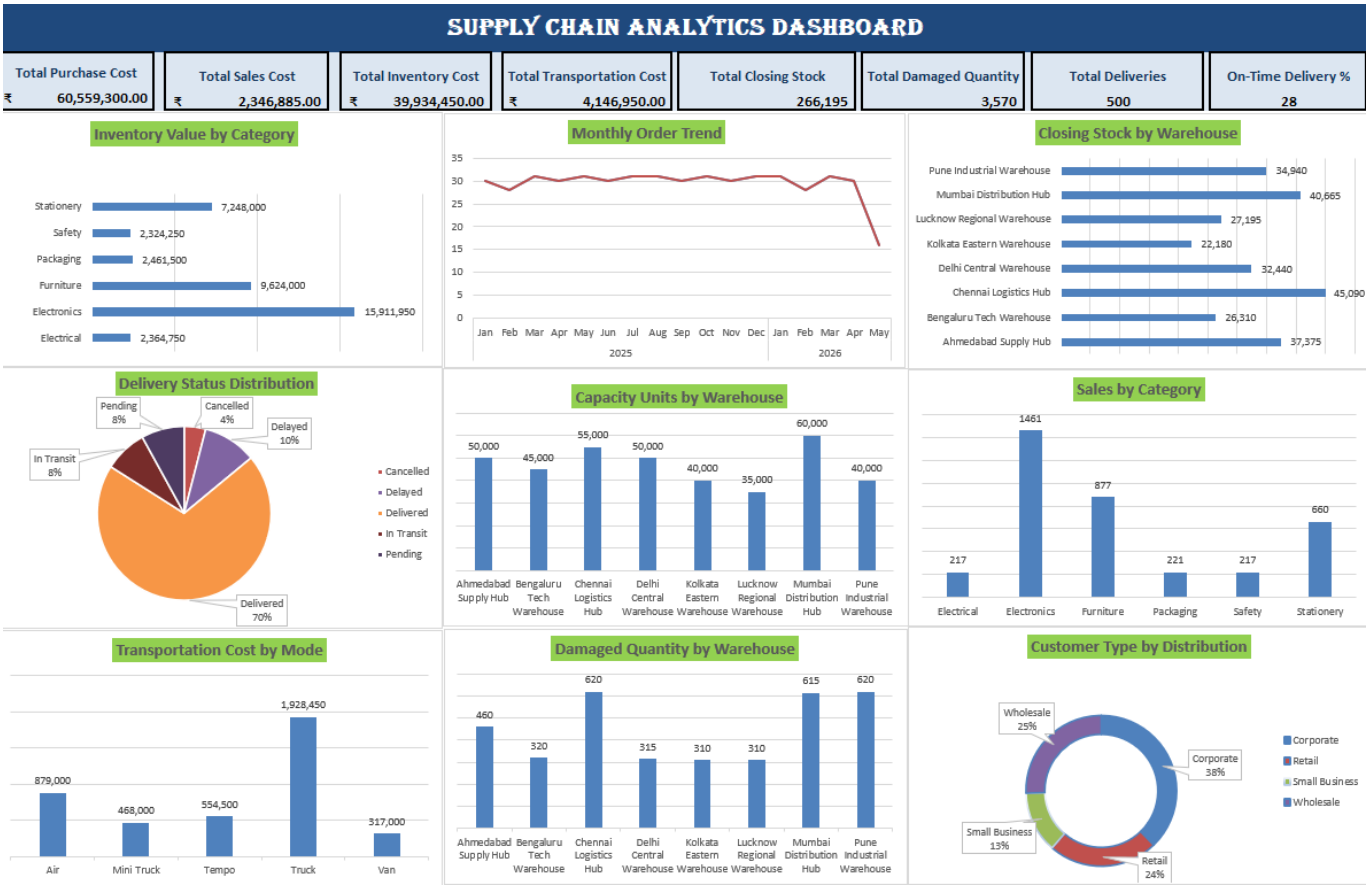
Figure C.1: Supply Chain Analytics Dashboard [Power BI]



Appendix D – Microsoft Excel Dashboard

This appendix presents the final dashboard developed using Microsoft Excel. It includes key performance indicators , charts, and analytical summaries related to supply chain operations.

Figure D.1: Final Supply Chain Analytics Dashboard [Microsoft Excel]



Appendix E – Python Dashboard and Visualizations

This appendix presents the dashboard and visualizations generated using Python. It includes analytical charts related to delivery status, customer distribution, product sales, transportation costs, and other supply chain activities.

Figure E.1: Supply Chain Analytics Dashboard [Python]

